

OVERALL PROJECT REPORT

Metro Ticket Generation System

ServiceNow • Phases 1–5 • Requirements to Final Presentation

Field	Detail
Platform	ServiceNow (custom scoped application, Service Catalog, Service Portal)
Delivery Structure	Five phases: Planning → Backend → UI/UX → Testing & Security → Deployment
Core Use Case	Passenger selects source and destination metro stations and generates a ticket with a dynamically calculated fare and a QR-ticket experience.

1. Project Overview & End-to-End Architecture

The Metro Ticket Generation System is developed in ServiceNow as a phased solution that moves from requirement analysis and planning through backend configuration, passenger-facing UI/UX, testing and security, and final deployment/documentation.

Five-Phase Delivery Structure

Phase	Primary Focus	Outcome
1	Requirement Analysis & Planning	Approved blueprint, scope, MVP, rules and roadmap
2	Backend Development & Configurations	Data layer, tables, fields, relationships and master data
3	UI/UX Development & Customization	Booking catalog, dynamic fare logic and QR experience
4	Data Migration, Testing & Security	Validation, QA, ACL review, data integrity and field mapping
5	Deployment, Documentation & Final Presentation	Troubleshooting, documentation, demo and future roadmap

Architecture

User → Service Catalog → Variables → Client/UI Logic → Validation → Automation / Field Mapping → Backend Table → Ticket / QR Experience

Phase 3 connects the passenger-facing catalog to the backend foundation. Phase 4 adds validation, security and field mapping, while Phase 5 prepares the solution for deployment, handoff and presentation.

2. Phase 1 — Requirement Analysis & Planning

Phase 1 establishes what the system must achieve before development begins. Its focus is requirement discovery, clarification, prioritization and planning. The planned journey is Passenger → Journey Details → Validation → Fare Calculation → Review → Confirmation → Ticket Generation → Notification → Ticket History.

MVP scope

- Station selection and source/destination validation
- Journey date and passenger-count validation
- Fare calculation
- Booking creation and unique ticket generation
- Ticket status and ticket viewing
- Confirmation notification

Out of initial scope

Real-time train tracking and external transit APIs, online payment gateway and financial settlement, QR gate scanning / smart-card integration, dedicated mobile application and advanced SMS/push integrations, and AI-based route optimization or dynamic pricing were identified as future/out-of-scope items.

Planning risks and roadmap

Key risks include unclear fare policy, changing requirements, duplicate tickets, incorrect access, poor master data and integration dependencies. The planning material proposes business approval, scope baselining, unique identifiers, early ACL definition, master-data validation and explicit MVP boundaries.

3. Phase 2 — Backend Development & Configurations

Phase 2 converts the Phase 1 requirements into a working ServiceNow backend structure: Metro Station Details, Metro Route, Metro Fare, Metro Booking, Metro Ticket and Passenger. The first concrete activity is creating the Metro Station Details table (u_metro_station_details) and its Station Name field (u_station_name), which anchors the rest of the data model.

Data Architecture → Tables → Fields → Relationships → Validation/Configuration → Backend Testing → Ready for Logic

Evidence — Metro Station Details Table Configuration

The screenshot shows the ServiceNow configuration page for the table 'u_metro_station_s_details'. The table is associated with the 'Global' application. The 'Columns' tab is active, showing a list of columns with the following details:

Column label	Type	Reference	Max length	Default value	Display
Journey Type	Choice	(empty)	40	false	false
Updates	Integer	(empty)	40	false	false
Station Name	String	(empty)	40	true	true
Destination	Reference	Metro Station's Details	32	false	false
Updated by	String	(empty)	40	false	false
Updated	Date/Time	(empty)	40	false	false
Total Fare	Integer	(empty)	40	false	false
Created by	String	(empty)	40	false	false
Source	Reference	Metro Station's Details	32	false	false
Created	Date/Time	(empty)	40	false	false
Sys ID	Sys ID (GUID)	(empty)	32	false	false

Metro Station Details table configuration showing the table definition and configured columns.

Evidence — Station Record Creation and Verification

Station Name	Journey Type	Source	Total Fare	Destination
Uppal Stadium		(empty)		(empty)
Madhapur		(empty)		(empty)
Jubilee Hills		(empty)		(empty)
Panjagutta		(empty)		(empty)
Kukatpally		(empty)		(empty)
Ameerpet		(empty)		(empty)
L B Nagar		(empty)		(empty)
(empty)	1	Ameerpet		Kukatpally
Booking Log 1	1	Jubilee Hills		Madhapur
(empty)	2	Ameerpet	200	Panjagutta
Booking Log 2	2	Panjagutta	200	Uppal Stadium

Metro Station Details list showing configured station records: Uppal Stadium, Panjagutta, Madhapur, L B Nagar, Kukatpally, Jubilee Hills and Ameerpet.

4. Phase 3 — UI/UX Development & Customization

Phase 3 builds the passenger-facing Service Catalog item Book A Metro Ticket, with eight variables (Starting From, Going To, Type of Journey, No. of Passengers, Amount for Single Journey, Amount Including Return, Mode of Payment, Enter Payment Mode), a Catalog Client Script for dynamic fare calculation, a Catalog UI Policy that reveals Enter Payment Mode only when Others is selected, and a second Catalog Client Script that opens a QR-ticket modal (metro_qr_widget) on the first Checkout click.

Route	Single / Passenger	Return / Passenger
Ameerpet → Madhapur	₹30	₹60
Ameerpet → LB Nagar	₹50	₹100
Ameerpet → Kukatpally	₹30	₹60
Ameerpet → Jubilee Hills	₹20	₹40

Evidence — Catalog Item Variables

dev228929.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D72303e8344310cf1d5d10feaad3c7%26sysparm_record_target%3Dsc_cat_item

servicenow All Favorites History Workspaces Admin Catalog Item - Book A Metro Ticket

Application scope: Global
Update set: Default (Global)

Meta

Copy Try It Update Edit in Catalog Builder Delete

Related Links
[Item Diagnostic](#)
[Show via render type](#)
[Run Point Scan](#)
[SN Utlb\Versions \(48\)](#)

Variables (8) Variable Sets Catalog UI Policies (3) Catalog Client Scripts (3) Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles Related Catalog Items Assigned Topics

Order Search

Catalog Item = Book A Metro Ticket

Type	Question	Order
Reference	Starting Form	100
Reference	Going To	200
Multiple Choice	Type of journey	300
Select Box	No of Passengers	400
Single Line Text	Amount for single journey	500
Single Line Text	Amount including return	600
Multiple Choice	Mode of Payment	700
Single Line Text	Enter payment mode	800

1 to 8 of 8

Maintain Items view showing the eight configured variables for Book A Metro Ticket.

Evidence — Book A Metro Ticket Configuration

dev228929.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D72303e8344310cf1d5d10feaad3c7%26sysparm_record_target%3Dsc_cat_item

servicenow All Favorites History Workspaces Admin Catalog Item - Book A Metro Ticket

Application scope: Global
Update set: Default (Global)

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Book A Metro Ticket

Application: Global

Active: ☒

Fulfillment automation level: Unspecified

Category: Services

State: -- None --

Checked out: -- None --

Owner: System Administrator

Secondary owners:

Item Details Process Engine Picture Pricing Portal Settings

Short description: A metro e-ticketing system

Description

Catalog Item form showing name, Service Catalog, category, application scope and short description.

Evidence — Service Portal Entry Point

Widget - Metro QR Widget

Name:

ID:

Description:

Application:

Public: ☐

Roles:

Body HTML template

```
<div style="text-align:center; padding:25px; font-family:Arial, sans-serif;">
<h3>Your Metro Ticket QR Code</h3>

<p>Scan this QR code at the metro station gate</p>
</div>
```

CSS

Service Portal landing page providing the passenger-facing interface.

Evidence — Passenger Booking Form

servicenow

Knowledge Catalog Requests System Status Cart Tours System Administrator

How can we help?

Request Something
Browse the catalog for services and items you need

Knowledge Base
Browse and search for articles, rate or submit feedback

Get Help
Contact support to make a request, or report a problem

Current Status
No system is reporting an issue
[More information...](#)

My Assessments and Surveys
No assessments or surveys for you at the moment

My Approvals
You have no pending approvals

Announcements
Employee Center is available to you

My Open Incidents
Employee payroll application server is down.
INC0007883 • 3mo ago
Rain is leaking on main DNS Server
INC0000814 • 3mo ago
JavaScript error on hiring page of corporate website
INC0000481 • 3mo ago
Can't launch 64-bit Windows 7 virtual machine
INC0000013 • 3mo ago

Top Rated Articles
Getting Around in Windows
★★★★★

Catalog form showing Starting From, Going To, journey type and passenger count.

Evidence — Fare and Payment Section

The screenshot shows the 'Book A Metro Ticket' form in the ServiceNow interface. The form is titled 'Book A Metro Ticket' and is described as 'A metro e-ticketing system'. It includes several input fields and a sidebar for additional options. The main form fields are: 'Starting Form' (Kukatpally), 'Going To' (Madhapur), 'Type of journey' (Single journey), 'No of Passengers' (1), 'Amount for single journey' (300), and 'Mode of Payment' (UPI). The sidebar on the right contains 'Quantity' (1), 'Delivery Time: 0 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. The bottom of the form has an 'Add attachments' section with a file upload icon and a link to 'Choose a file or drag it here. Copy and paste clipboard files here.'

Lower portion of the catalog form showing fare outputs, Enter Payment Mode and checkout controls.

Evidence — Metro QR Widget

This screenshot shows the 'Book A Metro Ticket' form in the ServiceNow interface, similar to the previous one but with a different fare output. The 'Amount including return' field is now set to 200. The sidebar on the right remains the same, showing 'Quantity' (1), 'Delivery Time: 0 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. The bottom of the form has an 'Add attachments' section with a file upload icon and a link to 'Choose a file or drag it here. Copy and paste clipboard files here.'

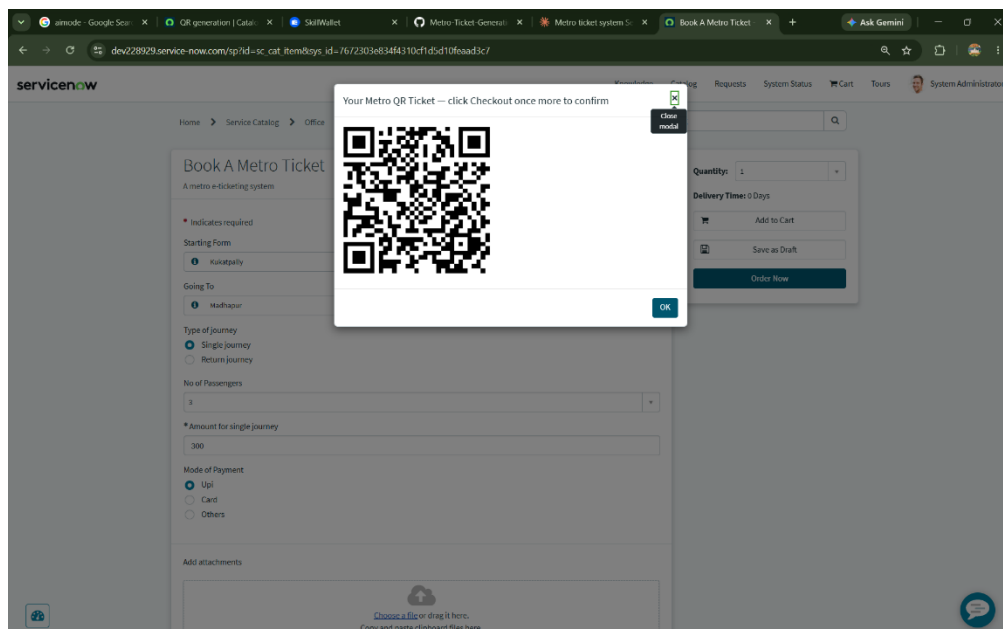
Widget record for Metro QR Widget, including widget ID and Body HTML template.

5. Phase 4 — Data Migration, Testing & Security

Phase 4 is the quality and security gate. Access is role-based: the default ACLs created with the custom table govern read, create, delete and write, so catalog access does not equal unrestricted database access. Functional, negative and boundary test cases confirm mandatory-field validation, fare accuracy, conditional payment fields and the QR checkout flow, while field mapping through the Process Automation Engine bridges catalog variables to backend fields.

Evidence — Calculated Fare and QR Ticket

Catalog form showing dynamically populated fare fields.



QR ticket modal displayed during the checkout flow.

6. Phase 5 — Deployment, Documentation & Final Presentation

Phase 5 closes the delivery lifecycle: structured troubleshooting for QR encoding, fare mismatches, flow execution and variable population; a ten-step setup manual for recreating the solution in a fresh PDI; a demo plan that tells a business story from portal walkthrough to QR ticket; and a future roadmap covering SMS/email ticketing, payment-gateway integration, an analytics dashboard, and a move toward data-driven, server-side fare calculation.

Evidence — Supplied Project Screenshots

MA Assigned to Manikanta Meesala

Description

- Debugged QR encoding issues and fare mismatches using logs and print statements in scripts.
- Flow **Execution Logs**: Used **Flow Designer's Execution Details** to trace flow steps, identify failures, and verify condition paths during approvals and notifications.
- Variable **Population Issues**: Resolved misaligned or missing data bindings using:
 - Catalog client scripts for dynamic population
- Form **Behaviour Testing**: Confirmed UI policies and visibility logic via Service Portal form tests and preview mode.

Resources

No resources available

Supplied implementation evidence.

MA Assigned to Manikanta Meesala

Description

- Implemented **end-to-end automation** using:
 - **Flow Designer** (no custom script includes or business rules)
 - **Dynamic Task creation**
 - **Client scripts and UI policies** kept to minimal, essential use
- Ensured maintainability and ease of handoff to admins or future developers.

Resources

No resources available

Supplied project evidence.

Evidence — Additional Visual and Presentation Evidence

MA Assigned to Manikanta Meesala

Description

- All setup steps annotated with screenshots, clear action-result mapping.
- Documentation and presentation follow **logical, sequential structure**:
 - What → Why → How → Outcome
- Each step includes:
 - Clear headings
 - Labelled screenshots
 - Explanatory notes for replicability.

Resources

No resources available

Supplied project evidence.

Project Demo Video Planning

Story

Duration: 5m

Review

MA

Assigned to Manikanta Meesala

Description

- Walkthrough from portal ticket submission to QR scan by station authority.
- Demo includes:
 - **Portal walkthrough:** Home → Book A Metro Services → Submit Request
 - **Live request submission**
 - **Request status tracking**
 - **Data Storing in Table.**
- Includes narration for each step and role-based perspectives (requester, fulfiller).

Resources

No resources available

Supplied project evidence.

7. Consolidated Exit Criteria Across Phases

Phase	Deliverable Confirmed
1	Business objectives, scope, MVP, business rules, data model and roadmap approved.
2	Metro Station Details table and Station Name field created; station records created and verified.
3	Book A Metro Ticket catalog item, variables, fare-calculation script, UI policy and QR widget working end-to-end.
4	ACL review, functional/negative/boundary QA, validation and field mapping completed.
5	Troubleshooting, setup manual, demo plan and future roadmap documented; solution deployment-ready.

8. Overall Project Outcome

Across the five phases, the Metro Ticket Generation System progressed from an approved requirements blueprint to a working ServiceNow application: a governed backend data layer, a dynamic Service Catalog booking experience with fare calculation and a QR-ticket presentation, role-based security enforced through ACLs, and a documented, reproducible and demonstrable final deliverable with a defined path for future scalability.

Final Delivery: Functional + Secure + Documented + Reproducible + Demonstrable + Scalable